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Optimal Operation of Data Center Microgrids Based on the Dynamic Surrogate Models of Server Power Consumption

Qian Chen, Shenxi Zhang, *Senior Member, IEEE*, Keyan Liu, Heng Zhang, Haozhong Cheng

***Abstract*—Internet data center microgrids (IDC-MGs) integrate data centers with renewable energy sources, energy storage and conversion devices to enhance sustainability. However, as the core load units of data centers, server power consumption exhibits significant temperature-dependent dynamics, posing significant challenges to the secure and economical operation of IDC-MGs. While existing mechanism models can accurately capture this dynamic behavior, their nonlinear and nonconvex nature precludes effective integration into system-level optimization frameworks. To address this challenge, this paper proposes a novel optimal operation method for IDC-MGs based on the dynamic surrogate model of server power consumption, enabling coordination between device-level dynamics and system-level energy management. Firstly, an input convex neural network (ICNN)-based dynamic surrogate model is developed to accurately capture the nonlinear dynamics of server power consumption, avoiding the complex mathematical expressions of mechanism models. Then, to reduce the model scale while maintaining computational accuracy, a novel response-driven temporal pruning (RDTP) method is proposed. This method effectively reduces the input temporal scope of the dynamic surrogate model by identifying the dynamic response boundaries of server power consumption. Finally, an optimal operation framework for IDC-MGs is developed, incorporating the dynamic surrogate model. This framework can jointly optimize energy device outputs and temporal shifting processes of information loads, while considering the dynamic characteristics of server power consumption. Case studies on both practical and test systems demonstrate the effectiveness of the proposed method. It achieves cost reductions of approximately 5.00% and 5.45%, respectively, compared to existing models that neglect the dynamic characteristics of server power consumption.**

***Index Terms*—data center microgrids; optimal operation; server power consumption; dynamic characteristics; input convex neural network; response-driven temporal pruning method**

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NOMENCLATURE

Acronyms

IDC-MG	Internet data center microgrid
ICNN	Input convex neural network
RDTP	Response-driven temporal pruning
WT	Wind turbine
PV	Photovoltaic
ESS	Energy storage system
MT	Microturbine
MEMS	Microgrid energy management system
PUE	Power usage efficiency
SOC	State of charge
MIQP	Mixed-integer quadratic program

Sets and indices

$\boldsymbol{\Gamma}$, t	Set and index of time slots
$\boldsymbol{\pi}$, τ	Set and index of time points for a single time slot
$\boldsymbol{\ell}$, l	Set and index of server clusters

Parameters

P_{other}	Other power consumption of servers
P_{idle}	Idle CPU power consumption of servers
a , b_0 , b_1	Parameters of CPUs
c_0^l , c_1^l , c_2^l , c_3^l , k_{fan}	Parameters of fans
T_{amb}	Ambient temperature
R_{cpu} , C_{cpu}	Equivalent thermal resistance and thermal capacity of CPUs
C_{hs}	Equivalent thermal capacity of heatsinks
R_{hs}^0	Equivalent thermal resistance of heatsinks at zero fan speed
d	Parameters of heatsinks
$f_{\text{fan}}^{\max}$	Maximum fan speed
a_{grid}	Unit electricity purchase price from the main grid
a_{MT} , b_{MT} , c_{MT}	Fuel cost coefficients of MTs
a_{penalty} , b_{penalty}	Penalty coefficients for tie-line power fluctuations
PUE	Power usage effectiveness of data centers

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L	Number of server clusters
$P_{\mathrm{TL}}^{\max}$, $P_{\mathrm{TL}}^{\min}$	Upper and lower limits of the tie-line transmission power
$P_{\mathrm{MT}}^{\max}$, $P_{\mathrm{MT}}^{\min}$	Upper and lower output limits of MTs
$R_{\mathrm{MT}}^{\max}$, $R_{\mathrm{MT}}^{\min}$	Ramp-up and ramp-down limits of MTs
E_{EES}	Storage capacity of EESs
η_{ch} , η_{dis}	Charge and discharge efficiencies of EESs
$SOC_{\max}$, $SOC_{\min}$	Upper and lower bounds of SOCs
$P_{\mathrm{EES}}^{\max}$	Maximum charge and discharge power
S_l	Allocation matrix for server cluster l
$\lambda_l^{\max}$	Maximum information load processing rate for server cluster l
G_l	Number of servers for server cluster l

Variables

$P_{\mathrm{ser}}^{t_\tau}$, $P_{\mathrm{cpu}}^{t_\tau}$, $P_{\mathrm{fan}}^{t_\tau}$	Total, CPU and cooling fan power consumption of servers at time t_τ
$P_{\mathrm{dyn}}^{t_\tau}$, $P_{\mathrm{leak}}^{t_\tau}$	Dynamic and leakage CPU power consumption of servers at time t_τ
$U_{\mathrm{cpu}}^{t_\tau}$	CPU utilization at time t_τ
$T_{\mathrm{cpu}}^{t_\tau}$	CPU core temperature at time t_τ
$f_{\mathrm{fan}}^{t_\tau}$	Fan speed at time t_τ
$R_{\mathrm{hs}}^{t_\tau}$	Equivalent thermal resistance of heatsinks at time t_τ
C_{total}	Daily operation cost of IDC-MGs
C_{grid} , C_{fuel} , C_{penalty}	Electricity purchase cost, fuel cost and tie-line power fluctuations penalty cost of IDC-MGs
P_{TL}^{t}	Tie-line transmission power at time t
P_{MT}^{t}	Power generation of MTs at time t
P_{WT}^{t} , P_{PV}^{t}	Output power of WTs and PVs at time t
$P_{\mathrm{ser,clu},l}^{t}$	Power consumption of the server cluster l at time t
$P_{\mathrm{EES,ch}}^{t}$, $P_{\mathrm{EES,dis}}^{t}$	Charge and discharge power of EESs at time t
SOC^{t}	SOC of EESs at time t
y^{t}	Charge and discharge state of EESs
$\lambda_l^{\mathrm{ar},t}$, $\lambda_{l,\mathrm{re}}^{\mathrm{ar},t}$, $\lambda_{l,\mathrm{tol}}^{\mathrm{ar},t}$	Total, delay-sensitive, and delay-tolerant information loads arriving rates for server cluster l at time t
$\lambda_{l,\mathrm{tol}}^{\mathrm{pro},t}$, $\lambda_l^{\mathrm{pro},t}$, $\lambda_{l,\mathrm{re}}^{\mathrm{pro},t}$	Total, delay-sensitive, and delay-tolerant information load processing rates for server cluster l at time t

I. Introduction

With the rapid development of IoT, cloud computing, and 5G technologies, data centers have become essential to modern information societies. However, their increasing energy consumption has emerged as a significant challenge for global energy and environmental systems [1]. To address the issue, internet data center microgrids (IDC-MGs) have been introduced as an innovative solution. IDC-MGs integrate data centers with distributed renewable energy sources, energy storage, and conversion devices. It can effectively reduce energy consumption and enhance energy self-sufficiency [2]. Moreover, optimizing the operation of IDC-MGs not only reduces the operation cost but also improves the overall system stability [3]. Consequently, investigating the optimal operation strategies for IDC-MGs has become a critical focus.

Several recent studies have focused on optimizing the operation of IDC-MGs. Ref. [4] addresses real-time energy management in IDC-MGs with a distributed approach. Ref. [5] proposes a novel optimization method based on demand response, while Ref. [6] explores an approach that integrates renewable energy and waste heat recovery. Additionally, Ref. [7] investigates a strategy that adjusts both the data center workload and UPS energy storage systems. Ref. [8] proposes a novel two-stage multi-energy sharing model considering joint grouping and matching priority. Ref. [9] investigates statistical feasibility robust optimization method for hydrogen-data center microgrid operations under uncertainty. Ref. [10] develops a two-stage robust planning method considering batch load flexibility and multi-energy complementarity. Since servers are the core load units of data centers, accurately modeling their power consumption is crucial for optimal system performance. This is critical for optimal IDC-MG operation, requiring real-time, high-accuracy coordination between data center loads and local energy resources. However, with the increasing CPU power density, temperature-dependent fluctuations in power consumption have become more pronounced. These fluctuations cause server power consumption to vary dynamically with CPU utilization due to thermal inertia [11]. This dynamic behavior complicates energy resource management and poses significant challenges to the secure, economical operation of IDC-MGs. However, existing studies often overlook this dynamic behavior. Therefore, it is essential to explore optimal operation methods that account for this dynamic to enhance operation efficiency and mitigate risks.

To facilitate the development of such methods, a key prerequisite is constructing server power consumption dynamic models that can be embedded into optimization problems. Existing models typically fall into two broad categories: static models that overlook the dynamic characteristics [12-14], and dynamic models that account for them [15-19]. Focusing on static models, Ref. [12] introduced a simple linear model (LM) based on no-load and full-load parameters. Ref. [13] proposed a piecewise linear model (PLM), while Ref. [14] employed a polynomial model (PM). While these models can be easily integrated into optimization frameworks, they fail to capture the dynamic characteristics of server power consumption, leading to significant deviations in calculation results. On the other hand, dynamic models, such as that in Ref. [15], include CPU temperature changes to improve the accuracy of power consumption calculations. Ref. [16] further split the CPU power consumption into dynamic

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and leakage components, offering detailed insights into how temperature affects power use. Ref. [17] investigated intelligent temperature-controlled cooling fans, emphasizing the relationship between the fan speed and measured temperature. Ref. [18] proposes an enhanced precision server power consumption model with temperature estimation based on CPU operating status, and Ref. [19] investigates energy-aware virtual machine placement based on a holistic thermal model for cloud data centers. Although these dynamic models provide accurate descriptions of server power consumption, their nonlinear and nonconvex nature pose challenges for the integration into optimization problems. Therefore, the development of models that are both accurate and easy to integrate remains a critical challenge.

Given the limitations of traditional modeling methods, machine learning-based approaches present a promising alternative. They eliminate the need for complex analysis of nonlinear mechanisms, simplifying high-dimensional nonlinear control challenges and offering substantial performance advantages. For example, Ref. [20] introduced a model that combines deep-belief networks with evolutionary algorithms to prevent and control transient instabilities. Ref. [21] applied extreme learning machines to construct an equivalent convex model for electricity pricing in bilateral auction markets. While these machine learning models provide superior computational efficiency, their non-convexity still poses a significant challenge when embedded within optimization frameworks. This non-convexity can produce local optima, rather than the global optimum [22]. To address the convexity requirement, convex regression methods have been developed for applications such as building energy forecasting and decarbonization pathway optimization [23, 24]. Similarly, the constrained Gaussian process, another approach to convex surrogate modeling, has been employed for probabilistic transient stability assessment and simulation-based optimization [25, 26]. However, these convex methods still face inherent trade-offs between dynamic modeling accuracy and optimization embeddability. To overcome these challenges, the input convex neural network (ICNN) model was introduced. The ICNN explicitly models the relationship between inputs and outputs as a nonlinear convex function, offering an innovative solution. By constructing compact convex safety boundaries, it avoids local optima and achieves excellent convergence efficiency [27], [28]. Notably, compared to traditional machine learning models, The ICNN provides rigorous convexity proof, making it a more reliable and efficient approach.

To address the aforementioned challenges, this paper proposes a novel optimal operation method for IDC-MGs based on the dynamic surrogate model of server power consumption, enabling coordination between device-level dynamics and system-level energy management. Firstly, an ICNN-based dynamic surrogate model is developed to accurately capture the nonlinear dynamics of server power consumption, avoiding the complex mathematical expressions of mechanism models. Then, to reduce the model scale while maintaining computational accuracy, a novel response-driven temporal pruning (RDTP) method is proposed. This method effectively reduces the input temporal scope of the dynamic surrogate model by identifying the dynamic response boundaries of server power consumption. Finally, an optimal operation framework for IDC-MGs is developed, incorporating the dynamic surrogate model. This framework can jointly optimize energy device outputs and temporal shifting processes of information loads, while considering the dynamic characteristics of server power consumption.

The main contributions of this paper are as follows:

- An ICNN-based dynamic surrogate model is developed to accurately capture the nonlinear dynamics of server power consumption, avoiding the complex mathematical expressions of mechanism models, and ensuring convexity.
- A novel RDTP method is proposed, which effectively reduces the input temporal scope of the dynamic surrogate model by identifying the dynamic response boundaries of server power consumption, reducing the model scale while maintaining computational accuracy.
- An IDC-MG optimal operation framework with the embedded dynamic surrogate model is established, which can jointly optimize energy device outputs and temporal shifting processes of information loads, while considering the dynamic characteristics of server power consumption.

The remainder of this paper is organized as follows. Section II presents the ICNN-based dynamic surrogate model for server power consumption. Section III introduces the RDTP method for the dynamic surrogate model. Section IV formulates the optimal operation framework for IDC-MGs. Case studies are discussed in Section V and VI, followed by discussions in Section VII and conclusions in Section VIII.

II. ICNN-Based Dynamic Surrogate Model for Server Power Consumption

This section first develops a server power consumption dynamic mechanism model, followed by the construction of the ICNN-based dynamic surrogate model.

A. *Server power consumption dynamic mechanism model*

As the core load units of data centers, servers play a pivotal role in modern computing ecosystems by executing complex data processing tasks, storing critical information, and providing essential computational resources for various applications. However, servers often consume significant power during operations, which inevitably generates heat, raising the temperature of components such as the CPU. This, in turn, introduces temperature-dependent dynamics and impacts the overall power consumption. This section analyzes the thermal behavior and power dynamics of servers based on the air-cooling technology architecture illustrated in Fig.1.

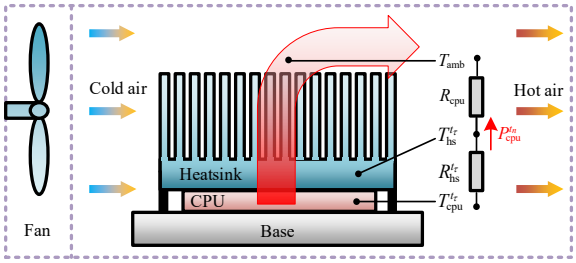
The overall power consumption of a server comprises three main components: CPU power consumption, cooling fan power consumption, and other power consumption [29]:

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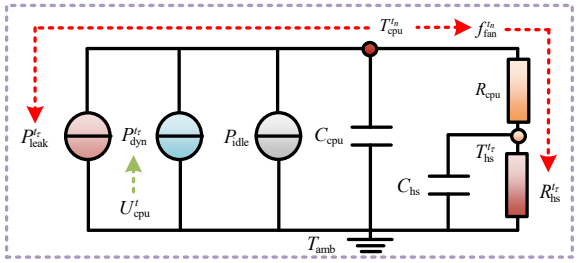
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$$P_{\text{ser}}^{t_\tau} = P_{\text{cpu}}^{t_\tau} + P_{\text{fan}}^{t_\tau} + P_{\text{other}}, \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \quad (1)$$

The CPU power consumption includes the idle power consumption, dynamic power consumption, and leakage power consumption [14]:



(a) Schematic diagram of the server structure and heat transfer process



(b) Schematic diagram of the equivalent thermal model inside the server

Fig. 1. Server architecture based on air-cooling technology

$$P_{\text{cpu}}^{t_\tau} = P_{\text{idle}} + P_{\text{dyn}}^{t_\tau} + P_{\text{leak}}^{t_\tau}, \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \quad (2)$$

The CPU dynamic power consumption arises from the transistor signal switching process and correlates with the computational tasks. It relates to the CPU utilization as follows [30]:

$$P_{\text{cpu}}^{t_\tau} = aU_{\text{cpu}}^{t_\tau}, \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \quad (3)$$

The CPU leakage power arises from the transistor leakage current, related to the internal carrier motion and temperature. It can be expressed as an exponential function of the CPU core temperature [18]:

$$P_{\text{leak}}^{t_\tau} = b_0 \exp\left(-\frac{b_1}{T_{\text{cpu}}^{t_\tau}}\right), \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \quad (4)$$

The power consumed by electronic components such as the CPU in a server is dissipated as thermal energy, raising the temperature. For cooling, modern servers often use temperature-controlled variable-speed fans, whose power consumption is modelled as follows [32]:

$$P_{\text{fan}}^{t_\tau} = c_0 + c_1 f_{\text{fan}}^{t_\tau} + c_2 \left(f_{\text{fan}}^{t_\tau}\right)^2 + c_3 \left(f_{\text{fan}}^{t_\tau}\right)^3, \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \quad (5)$$

The fan speed is related to the difference between the CPU and ambient temperatures [14]:

$$f_{\text{fan}}^{t_\tau} = k_{\text{fan}}\left(T_{\text{cpu}}^{t_\tau} - T_{\text{amb}}\right), \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \quad (6)$$

As shown in Fig. 1(b), the thermodynamic model of the server is expressed as follows [33]:

$$C_{\text{cpu}} \frac{dT_{\text{cpu}}^{t_\tau}}{dt} = P_{\text{cpu}}^{t_\tau} - \frac{T_{\text{cpu}}^{t_\tau} - T_{\text{hs}}^{t_\tau}}{R_{\text{cpu}}}, \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \quad (7)$$

$$C_{\text{hs}} \frac{dT_{\text{hs}}^{t_\tau}}{dt} = \frac{T_{\text{cpu}}^{t_\tau} - T_{\text{hs}}^{t_\tau}}{R_{\text{cpu}}} - \frac{T_{\text{hs}}^{t_\tau} - T_{\text{amb}}}{R_{\text{hs}}^{t_\tau}}, \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \quad (8)$$

The equivalent thermal resistance of the heatsink depends on the fan speed as follows [18]:

$$R_{\text{hs}}^{t_\tau} = \frac{r_{\text{hs}}^0}{d \bullet \left(f_{\text{fan}}^{t_\tau} / f_{\text{fan}}^{\max}\right)^2 + 1}, \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \quad (9)$$

Fig. 2 depicts the dynamic response characteristics of the CPU core temperature and server power consumption under different operating states. Specifically, when server utilization undergoes a step change, the power consumption exhibits a gradual adjustment toward a new equilibrium point, mediated by the thermal regulation mechanism. In fluctuation scenarios, the power consumption exhibits periodic oscillations around a stabilized range. This behavior is driven by the delay in heat transfer, where temperature changes occur after power fluctuations, leading to a dynamic adjustment in server power consumption.

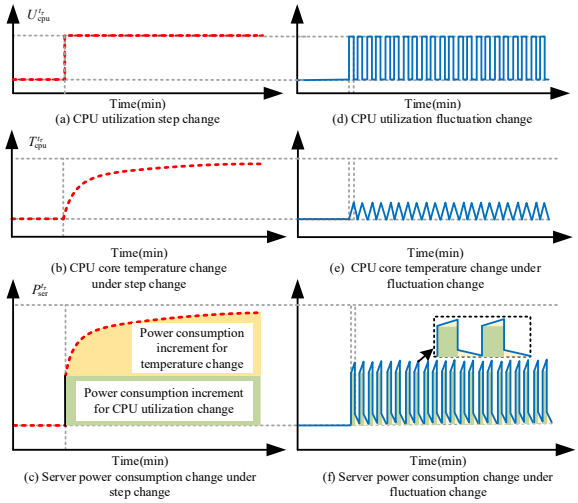


Fig. 2. Schematic diagram of parameter changes under different CPU operating states

Table I further analyzes the dynamic response time of server power consumption, which typically ranges from 4 to 12 minutes, matching the time step of IDC-MG optimization. However, current research overlooks this factor, sacrificing both model accuracy and solution optimality. Therefore, it is crucial to account for the dynamic characteristics of server power consumption in IDC-MG optimal operation.

TABLE I SERVER POWER CONSUMPTION DYNAMIC RESPONSE TIME

Reference	Time constant	Full response time
Ref. [16]	1.5~3 min	4~12 min
Ref. [18]	2.5 min	7~10 min
Ref. [30]	-	4~6 min
Ref. [33]	-	6~10 min
Ref. [34]	-	4~5 min

B. *Dynamic surrogate modeling based on ICNN*

The server power consumption dynamic mechanism model formulated by (1)-(9) incorporates both differential and nonlinear terms, posing significant challenges for its direct integration into the IDC-MG optimal operation problem. To

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address this limitation, this study develops an ICNN-based dynamic surrogate model.

The ICNN model can be represented as follows [28].

$$\boldsymbol{Z}_k = \boldsymbol{W}_k \boldsymbol{E}_{k-1} + \boldsymbol{D}_k \boldsymbol{X} + \boldsymbol{b}_k, \boldsymbol{W}_k \geq 0 \tag{10}$$

$$\boldsymbol{E}_k = g\left(\boldsymbol{Z}_k\right) \tag{11}$$

where $\boldsymbol{Z}_k = \left[z_k^1, z_k^2, ..., z_k^O\right]$ represents the input matrix of layer $k = 1, ..., M$; $\boldsymbol{E}_k = \left[e_k^1, e_k^2, ..., e_k^O\right]$ is the output matrix of layer k; $\boldsymbol{W}_k$ and $\boldsymbol{b}_k$ denote the weight and bias matrices of layer k, respectively; $\boldsymbol{D}_k$ represents the direct connection matrix between layers $k-1$ and k (with $\boldsymbol{D}_1 = \boldsymbol{0}$); $g(\bullet)$ is the activation function.

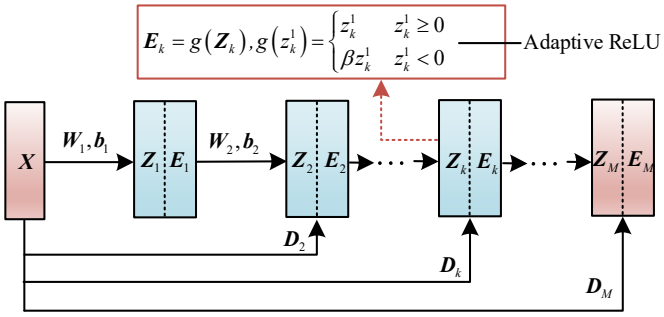
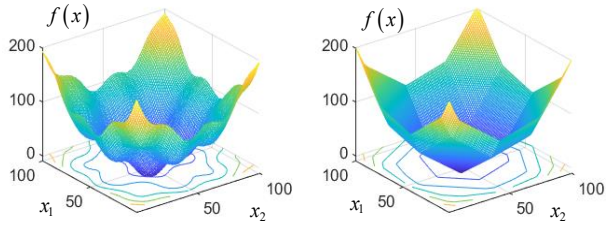


Fig. 3. Structure of the ICNN model



(a) Validation function ouput (b) ICNN output

Fig. 4. Approximation of function

To mitigate the dead neuron issue of standard ReLU and better preserve the convexity of ICNN, we use the adaptive ReLU as the activation function in ICNN, mathematically formulated as follows:

$$g\left(z_k^i\right) = \begin{cases} z_k^i & z_k^i \geq 0 \\ \beta z_k^i & z_k^i < 0 \end{cases} \tag{12}$$

where β is an adaptive parameter.

To evaluate the convex approximation capability of the ICNN, Fig. 4 compares its performance against a synthetically designed multimodal nonconvex surface:

$$y = x_1^2 + x_2^2 + 10\sin\left(x_1\right) + 10\sin\left(x_2\right) + x_3 \tag{13}$$

where x_1 and x_2 are the convex input variables; x_3 represents the non-convex noise component.

The training dataset was generated using (13) with uniformly distributed noise added. A three-layer ICNN architecture was adopted for function approximation, featuring hidden layers configured to [32, 32, 32] and adaptive ReLU activation functions applied to the hidden neurons. Fig. 4(a) illustrates the deliberately designed non-convex surface landscape containing multiple local minima. In contrast, the ICNN-approximated surface shown in Fig. 4(b) demonstrates rigorous convexification: it eliminates all original local minima while preserving convexity constraints. This example empirically verifies the convexity-enforcing capability of the ICNN.

Leveraging the inherent advantages of ICNN in preserving convexity, we apply it to develop a dynamic surrogate model for server power consumption, aiming to establish a convex mapping between the server utilization and power consumption. However, due to the temporal coupling in the model, the input-output settings require consideration of time dependencies, specifically:

$$\boldsymbol{X} = \left[U_{\text{cpu}}^{t_1}, ..., U_{\text{cpu}}^{t_n}, ..., U_{\text{cpu}}^{t_N}, ..., U_{\text{cpu}}^{t_N}\right], \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \tag{14}$$

$$\boldsymbol{Y} = \left[P_{\text{ser}}^{t_1}, ..., P_{\text{ser}}^{t_n}, ..., P_{\text{ser}}^{t_N}\right], \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \tag{15}$$

where N denotes the number of time points for a single time slot.

Table II compares three convex modeling methods, highlighting the distinct advantages of the ICNN for this work. While convex regression ensures structural convexity, it suffers from limited expressiveness in dynamic modeling [23, 24]. This limitation arises from its reliance on overly simplistic basis functions, such as piecewise linear ones, which lack the expressive ability to capture complex server power consumption dynamics. Constrained Gaussian processes provide high dynamic modeling accuracy and uncertainty quantification [25, 26]. However, their probabilistic, non-convex posterior form inherently hinders optimization applications, as it is not readily convertible into tractable, deterministic convex constraints. In contrast, the ICNN uniquely satisfies the core requirements for this application. It achieves high-accuracy modeling of dynamic server power consumption through powerful, parameterized nonlinear neuron bases. Furthermore, its architectural design guarantees a deterministically convex mapping. This structural convexity allows the model to be seamlessly embedded as tractable mixed-integer linear programming constraints via exact linearization. Consequently, the ICNN is chosen as the foundation for the dynamic surrogate model, effectively bridging the gap between dynamic modeling accuracy and optimization embeddability.

TABLE II COMPARISON OF CONVEX MODELING METHODS

Method	Dynamic modeling accuracy	Optimization embeddability
Convex regression	Low	Moderate
Constrained Gaussian process	High	Low
ICNN	High	High

III. RESPONSE-DRIVEN TEMPORAL PRUNING METHOD FOR THE DYNAMIC SURROGATE MODEL

The server power consumption dynamic surrogate model is characterized by (10)-(12), (14), and (15). It incorporates historical data inputs, which results in a large-scale model. Consequently, embedding this model into an optimization

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problem may reduce computational efficiency. To address this, the RDTP method is proposed. This approach reduces the model scale while maintaining accuracy, enhancing the practicality of the dynamic surrogate model.

To clarify, the maximum time constant for server power consumption is formally defined. This parameter quantifies the time interval required for a CPU utilization change to fully propagate and manifest as a power consumption response. Once determined, it provides a critical reference for specifying the input temporal scope of the dynamic surrogate model.

The maximum time constant is derived using the state-space method. It calculates the sensitivity matrix to capture the linear relationship between all inputs and outputs, thereby identifying the range of input influence. The dynamic characteristics of server power consumption are governed by state variables, and the nonlinear dynamic behavior of the server is expressed as follows:

$$\begin{cases} \dot{\boldsymbol{x}}^{t_\tau} = f\left(\boldsymbol{x}^{t_\tau}, U_{\text{cpu}}^{t_\tau}\right), \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \\ P_{\text{ser}}^{t_\tau} = h\left(\boldsymbol{x}^{t_\tau}, U_{\text{cpu}}^{t_\tau}\right), \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \end{cases} \tag{16}$$

where $\boldsymbol{x}^{t_\tau} = \left[T_{\text{cpu}}^{t_\tau} \quad T_{\text{hs}}^{t_\tau} \right]$ represents the state variables; $f(\bullet)$ and $h(\bullet)$ are functions containing the temperature vector derived from (1) to (9).

Near the steady-state point $(\boldsymbol{x}_0, U_0)$, a first-order Taylor expansion is applied to the nonlinear state equations:

$$\Delta\dot{\boldsymbol{x}}^{t_\tau} = \boldsymbol{A}\Delta\boldsymbol{x}^{t_\tau} + \boldsymbol{B}\Delta U_{\text{cpu}}^{t_\tau}, \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \tag{17}$$

where $\boldsymbol{A}$ and $\boldsymbol{B}$ are the input and state Jacobian matrices, respectively, computed as:

$$\boldsymbol{A} = \left.\frac{\partial f}{\partial \boldsymbol{x}}\right|_{(\boldsymbol{x}_0, U_0)}, \boldsymbol{B} = \left.\frac{\partial f}{\partial u}\right|_{(\boldsymbol{x}_0, U_0)} \tag{18}$$

The output equation is linearized as follows:

$$\Delta P_{\text{ser}}^{t_\tau} = \boldsymbol{C}\Delta\boldsymbol{x}^{t_\tau} + D\Delta U_{\text{cpu}}^{t_\tau}, \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \tag{19}$$

where $\boldsymbol{C}$ is the output Jacobian matrix; D is the direct transmission value, both computed as:

$$\boldsymbol{C} = \left.\frac{\partial h}{\partial \boldsymbol{x}}\right|_{(\boldsymbol{x}_0, U_0)}, D = \left.\frac{\partial h}{\partial u}\right|_{(\boldsymbol{x}_0, U_0)} \tag{20}$$

Using the Euler method, the continuous-time model is discretized:

$$\Delta\boldsymbol{x}^{t_\tau} = \left(\boldsymbol{I} - \Delta\tau \cdot \boldsymbol{A}\right)^{-1} \Delta\boldsymbol{x}^{t_{\tau-1}} + \Delta\tau\left(\boldsymbol{I} - \Delta\tau \cdot \boldsymbol{A}\right)^{-1} \boldsymbol{B}\Delta U_{\text{cpu}}^{t_\tau}, \quad \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \tag{21}$$

where $\Delta\tau$ is the discrete time step; $\boldsymbol{I}$ is the identity matrix.

Recursive application of (21) yields:

$$\Delta\boldsymbol{x}^{t_\tau} = \boldsymbol{F}^{(N-1)\cdot t+\tau}\Delta\boldsymbol{x}^{1_0} + \sum_{i_j=1_0}^{t_{\tau-1}} \boldsymbol{F}^{(N-i)\cdot t+(\tau-j)}\Delta\tau\boldsymbol{B}\Delta U_{\text{cpu}}^{t_{j+1}}, \quad \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \tag{22}$$

Hence, the recursive formula for the output equation is derived:

$$\Delta P_{\text{ser}}^{t_\tau} = \boldsymbol{C}\boldsymbol{F}^{(N-1)\cdot t+\tau}\Delta\boldsymbol{x}^{1_0} + \sum_{i_j=1_0}^{t_{\tau-1}} \boldsymbol{C}\boldsymbol{F}^{(N-i)\cdot t+(\tau-j)}\Delta\tau\boldsymbol{B}\Delta U_{\text{cpu}}^{t_{j+1}} + D\Delta U_{\text{cpu}}^{t_\tau}, \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \tag{23}$$

where $\boldsymbol{F} = \left(\boldsymbol{I} + \Delta\tau \cdot \boldsymbol{A}\right)^{-1}$ is the core operator of the state-space recursion.

Based on (23), the sensitivity matrix relating CPU utilization and server total power consumption across all time periods is constructed as follows:

$$\begin{bmatrix} \Delta P_{\text{ser}}^{1_1} \\ \Delta P_{\text{ser}}^{1_2} \\ \vdots \\ \Delta P_{\text{ser}}^{T_N} \end{bmatrix} = \begin{bmatrix} h_1 & 0 & \cdots & 0 \\ h_2 & h_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ h_{TN} & h_{TN-1} & \cdots & h_1 \end{bmatrix} \begin{bmatrix} \Delta U_{\text{cpu}}^{1_1} \\ \Delta U_{\text{cpu}}^{1_2} \\ \vdots \\ \Delta U_{\text{cpu}}^{T_N} \end{bmatrix} + D \begin{bmatrix} \Delta U_{\text{cpu}}^{1_1} \\ \Delta U_{\text{cpu}}^{1_2} \\ \vdots \\ \Delta U_{\text{cpu}}^{T_N} \end{bmatrix} + \boldsymbol{E}, \tag{24}$$

$$\forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi}$$

$$h_k = \boldsymbol{C}\boldsymbol{F}^k\Delta\tau\boldsymbol{B}, \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \tag{25}$$

$$\boldsymbol{E} = \left[\boldsymbol{C}\boldsymbol{F}^1\Delta\boldsymbol{x}^{1_0} \quad \boldsymbol{C}\boldsymbol{F}^2\Delta\boldsymbol{x}^{1_0} \quad \cdots \quad \boldsymbol{C}\boldsymbol{F}^{TN}\Delta\boldsymbol{x}^{1_0} \right], \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \tag{26}$$

where h_k is the parameter of the sensitivity matrix; $\boldsymbol{C}^{t_\tau}$ is the parameter matrix; T denotes the number of time slots.

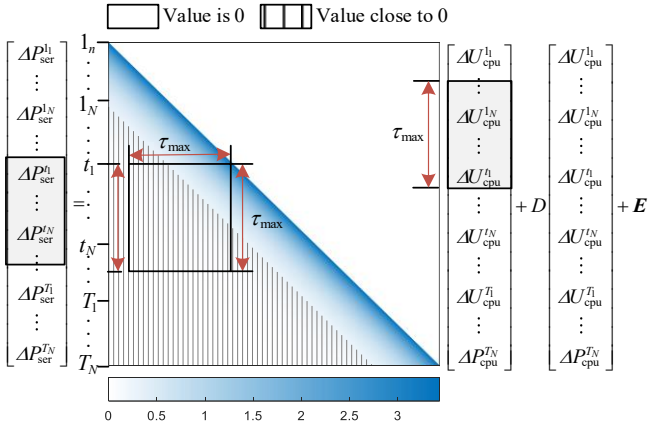


Fig. 5. Values of the sensitivity matrix

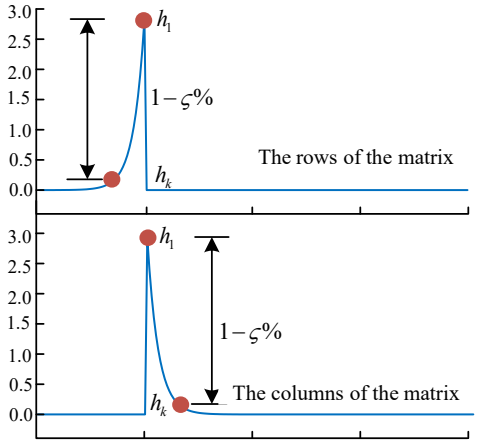


Fig. 6. Numerical curve for rows and columns of the sensitivity matrix

The sensitivity matrix represents the linear relationship between the input and output variations. Fig. 5 shows its values, with higher magnitudes near the diagonal and lower ones at the lower-left corner, indicating decreasing input influence over time. To determine the maximum time constant, a submatrix is extracted. The horizontal axis represents the influence of historical CPU utilization on server power consumption, while the vertical axis shows the impact of current CPU utilization on future server power consumption.

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Fig. 6 presents the numerical curve for the rows and columns of the sensitivity matrix.

The maximum time constant is calculated as:

$$\tau_{\max} = (k-1)\Delta\tau, if \ \frac{h_k}{h_1} \times 100\% \le \varsigma\% \tag{27}$$

where $\varsigma\%$ is the threshold ratio. It quantifies the relative influence of historical CPU utilization versus the current CPU utilization on the server power consumption.

After determining the maximum time constant via (27), the input temporal scope for ICNN is set as $\tau_{\max} + \Delta t$. Consequently, the ICNN input variables are:

$$\boldsymbol{X} = \left[\underbrace{U_{\text{cpu}}^{l,s_p},...,U_{\text{cpu}}^{(t-1)_N}}_{\tau_{\max}}, \underbrace{U_{\text{cpu}}^{t_1},...,U_{\text{cpu}}^{t_N}}_{\Delta t} \right], \forall t \in \boldsymbol{\Gamma}, \forall \tau \in \boldsymbol{\pi} \tag{28}$$

where Δt denotes the time step of the optimization framework. The dynamic surrogate model formed by (10)-(12), (15), and (28), reduces model scale while maintaining computational accuracy, enhancing its applicability in optimization problems.

To clarify the fundamental distinctions between the proposed method and standard truncation methods, a systematic comparison is presented in Table III. While standard truncation [35] is inherently empirical and arbitrary in nature, the proposed RDTP method objectively determines the input temporal scope. This is achieved through theoretical derivation, which identifies the dynamic response boundaries. By doing so, RDTP ensures that all essential historical data within these boundaries is fully utilized to maintain accuracy. Simultaneously, all data beyond the boundaries is discarded to avoid redundancy. Consequently, RDTP provides a superior foundation for constructing compact and efficient surrogate models, making them ideally suited for subsequent optimization problems.

TABLE III COMPARISON BETWEEN THE RDTP AND STANDARD TRUNCATION METHODS

Method	Input temporal scope	Dynamic accuracy preservation
Standard truncation	Empirical	Arbitrary
RDTP	Theoretical derivation	Optimal

IV. IDC-MG OPTIMAL OPERATION FRAMEWORK WITH THE EMBEDDED DYNAMIC SURROGATE MODEL

This section establishes an optimal operation framework for IDC-MGs with the embedded dynamic surrogate model. Fig. 7 illustrates the IDC-MG system architecture and optimal operation framework. The IDC-MG integrates a data center (including multiple server clusters) with wind turbines (WTs), photovoltaics (PVs), energy storage systems (ESSs), and microturbines (MTs). The system operates in grid-connected mode, enabling electricity trading with the main grid. The optimal operation is managed by the microgrid energy management system (MEMS). The framework calculates server power consumption at a second-level time step using the dynamic surrogate model. It generates time-varying constraint boundaries and solves the optimization problem at a minute-level time step, enabling coordinated power outputs of multiple energy devices.

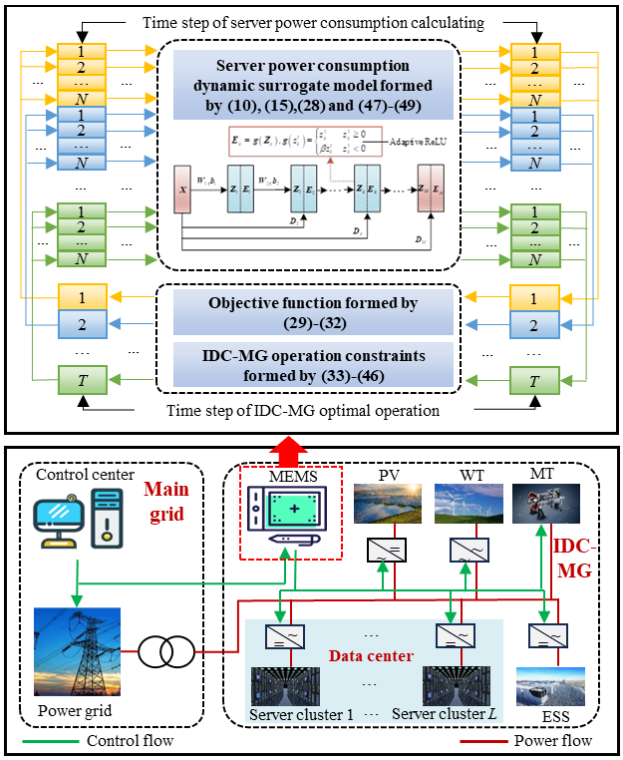


Fig. 7. IDC-MG system architecture and optimal operation framework

A. *Objective function*

The objective of IDC-MGs optimal operation is to minimize the daily operation cost C_{total}, comprising the electricity purchase cost C_{grid}, the fuel cost C_{fuel}, and the penalty cost for tie-line power fluctuations C_{penalty}. The cost expressions are:

$$\min C_{\text{total}} = C_{\text{grid}} + C_{\text{fuel}} + C_{\text{penalty}} \tag{29}$$

$$C_{\text{grid}} = \sum_{t=1}^{T} a_{\text{grid}} P_{\text{TL}}^{t} \tag{30}$$

$$C_{\text{fuel}} = \sum_{t=1}^{T} \left(a_{\text{MT}} \left(P_{\text{MT}}^{t} \right)^2 + b_{\text{MT}} P_{\text{MT}}^{t} + c_{\text{MT}} \right) \tag{31}$$

$$C_{\text{penalty}} = \sum_{t=1}^{T} a_{\text{penalty}} \left(P_{\text{TL}}^{t} - P_{\text{TL}}^{t-1} \right)^2 + b_{\text{penalty}} \left(P_{\text{TL}}^{t} - P_{\text{TL}}^{t-1} \right) \tag{32}$$

B. *Constraints*

The power balance constraints are:

$$P_{\text{TL}}^{t} + P_{\text{WT}}^{t} + P_{\text{PV}}^{t} + P_{\text{MT}}^{t} + P_{\text{EES,dis}}^{t} = PUE \bullet \sum_{l=1}^{L} P_{\text{ser,clu},l}^{t} + P_{\text{EES,ch}}^{t}, \tag{33}$$

$$\forall t \in \boldsymbol{\Gamma}, \forall l \in \ell$$

Tie-line transmission power can be expressed as follows:

$$P_{\text{TL}}^{\min} \le P_{\text{TL}}^{t} \le P_{\text{TL}}^{\max}, \forall t \in \boldsymbol{\Gamma} \tag{34}$$

The MT operation constraints include the output and ramping limits:

$$P_{\text{MT}}^{\min} \le P_{\text{MT}}^{t} \le P_{\text{MT}}^{\max}, \forall t \in \boldsymbol{\Gamma} \tag{35}$$

$$R_{\text{MT}}^{\min} \le P_{\text{MT}}^{t} - P_{\text{MT}}^{t-1} \le R_{\text{MT}}^{\max}, \forall t \in \boldsymbol{\Gamma} \tag{36}$$

The energy storage constraints are:

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$$SOC^{t+1}=\frac{SOC^{t}+\left(P^{t}_{\mathrm{EES,ch}}\eta_{\mathrm{ch}}-P^{t}_{\mathrm{EES,dis}}/\eta_{\mathrm{dis}}\right)\Delta t}{E_{\mathrm{EES}}},\forall t\in\boldsymbol{\Gamma} \quad (37)$$

$$SOC_{\min}\le SOC^{t}\le SOC_{\max},\forall t\in\boldsymbol{\Gamma} \quad (38)$$

$$0\le P^{t}_{\mathrm{EES,ch}}\le y^{\mathrm{t}}P^{\max}_{\mathrm{EES}},\forall t\in\boldsymbol{\Gamma} \quad (39)$$

$$0\le P^{t}_{\mathrm{EES,dis}}\le\left(1-y^{\mathrm{t}}\right)P^{\max}_{\mathrm{EES}},\forall t\in\boldsymbol{\Gamma} \quad (40)$$

$$SOC^{1}=SOC^{T+1} \quad (41)$$

The data center information load comprises delay-tolerant and delay-sensitive types. The delay-tolerant load migrates temporally:

$$\lambda_{l}^{\mathrm{ar},t}=\lambda_{l,\mathrm{re}}^{\mathrm{ar},t}+\lambda_{l,\mathrm{tol}}^{\mathrm{ar},t},\forall t\in\boldsymbol{\Gamma},\forall l\in\boldsymbol{\ell} \quad (42)$$

The maximum delay time period for the delay-tolerant load is denoted as R, and its allocation satisfies:

$$S_{l}=\begin{bmatrix}\lambda_{l}^{11} & \cdots & \lambda_{l,\mathrm{tol}}^{1R} & 0 & 0 & \cdots & 0\\ 0 & \lambda_{l,\mathrm{tol}}^{22} & \cdots & \lambda_{l,\mathrm{tol}}^{2,R+1} & 0 & \cdots & 0\\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \cdots & 0 & 0 & \cdots & \lambda_{l,\mathrm{tol}}^{NN}\end{bmatrix}_{0}, \quad (43)$$

$$\lambda_{k,\mathrm{tol}}^{ii}\ge 0,t\in\boldsymbol{\Gamma},\forall l\in\boldsymbol{\ell}$$

$$\lambda_{l,\mathrm{tol}}^{\mathrm{ar},t}=\sum_{i=1}^{N}S_{l,\mathrm{tol}}^{ti},\lambda_{l,\mathrm{tol}}^{\mathrm{pro},t}=\sum_{i=1}^{N}S_{l,\mathrm{tol}}^{it},\forall t\in\boldsymbol{\Gamma},\forall l\in\boldsymbol{\ell} \quad (44)$$

The information load processing rate is:

$$\lambda_{l}^{\mathrm{pro},t}=\lambda_{l,\mathrm{re}}^{\mathrm{pro},t}+\lambda_{l,\mathrm{tol}}^{\mathrm{pro},t},\forall t\in\boldsymbol{\Gamma},\forall l\in\boldsymbol{\ell} \quad (45)$$

The information load processing rate relates to server utilization as:

$$U^{t}_{\mathrm{cpu},l}=\frac{\lambda_{l}^{\mathrm{pro},t}}{G_{l}\lambda_{l}^{\max}},\forall t\in\boldsymbol{\Gamma},\forall l\in\boldsymbol{\ell} \quad (46)$$

Additionally, the model includes the dynamic surrogate constraints (10)-(12), (15), and (28). Together with (46), they constitute the complete model that connects the information load to the server power consumption.

The nonlinearity of (11)-(12) complicates optimization. Thus, the Big-M method linearizes them:

$$M\left(u_{k}^{i}-1\right)\le z_{k}^{i}\le Mu_{k}^{i} \quad (47)$$

$$z_{k}^{i}-M\left(1-u_{k}^{i}\right)\le e_{k}^{i}\le z_{k}^{i}+M\left(1-u_{k}^{i}\right) \quad (48)$$

$$\beta z_{k}^{i}-Mu_{k}^{i}\le e_{k}^{i}\le\beta z_{k}^{i}+Mu_{k}^{i} \quad (49)$$

To address the time-step mismatch between the dynamic surrogate model (second-level) and the optimization model (minute-level), the following conversion is applied:

$$U^{t_{\tau}}_{\mathrm{cpu},l}=U^{t}_{\mathrm{cpu},l},P^{t}_{\mathrm{ser,clu},l}=\frac{1}{N}\sum_{\tau=1}^{N}G_{l}P^{t_{\tau}}_{\mathrm{ser},l},\forall t\in\boldsymbol{\Gamma},\forall\tau\in\boldsymbol{\pi},\forall l\in\boldsymbol{\ell} \quad (50)$$

The optimal operation model is formulated as a mixed-integer quadratic program (MIQP), characterized by the inclusion of integer variables and convex quadratic penalty terms. Although the integer variables render the overall model nonconvex, its continuous relaxation is a convex quadratic program. In this relaxed problem, all continuous components remain convex. This specific structure classifies the problem as a convex MIQP, a class for which the global optimal solution can be theoretically guaranteed. The structural convexity of the ICNN surrogate model is the key enabler for this formulation. In contrast, a non-convex surrogate model would lead to a general non-convex MIQP, which offers no guarantee of global optimality.

V. Primary Case Study

A. *Practical system description*

The practical system in southern China, shown in Fig. 8, is adapted for optimal operation simulations. The system includes a WT with a 10 MW capacity, a PV with a 5 MW capacity, an MT with a 2 MW capacity and ramp rates of 0.4 MW/15 min, and an EES with a 4 MWh capacity and 1.3 MW maximum power. The data center comprises three server clusters: Server cluster 1 has 5,000 high-performance servers (Dell PowerEdge R750, Server 1), each consuming approximately 720W; Server cluster 2 has 10,000 cloud servers (HPE ProLiant DL320 Gen10 Plus, Server 2), each consuming around 270W; and Server cluster 3 has 15,000 edge servers (HPE ProLiant DL320 Gen10 Plus, Server 3), each consuming around 110W. The server parameters are listed in Table IV, and the data center operates at a PUE of 1.4.

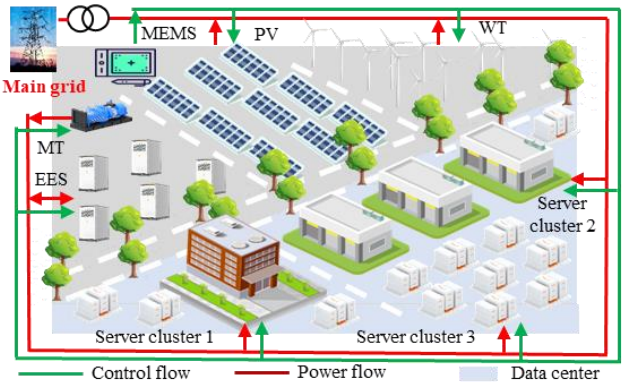


Fig. 8. Schematic diagram of the practical test system

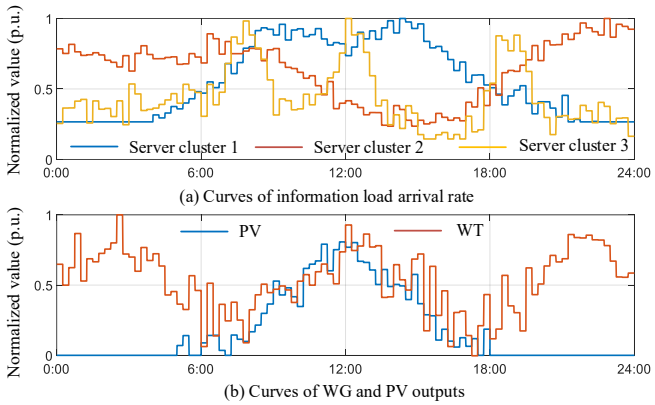
The upper and lower limits of the tie-line transmission power are -2 MW and 2 MW, respectively. The electricity purchase price is 0.7 yuan/kWh, and the fuel cost coefficients for the MT are $a_{\mathrm{MT}}=0.0001$, $b_{\mathrm{MT}}=0.5$, and $c_{\mathrm{MT}}=200$. The power fluctuation penalty coefficients are $a_{\mathrm{penalty}}=0.0001$ and $b_{\mathrm{penalty}}=0.05$. The normalized arrival rate of information load, and output of WG & PV are shown in Fig. 9. For information loads, the real-time to delay-tolerant load ratio is 1:1, with a maximum delay for R=1 hour for delay-tolerant loads. The maximum processing rates for Servers 1, 2, and 3 are 15, 5, and 2 request/s, respectively. The peak information load arrival rates for Server clusters 1, 2, and 3 are 70326, 31797, and 24222 request/s, respectively. The server utilization rates are constrained to a lower limit of 0.1 and an upper limit of 0.95. The time step of the optimization operation is $\Delta t=15\mathrm{min}$,The system simulations are conducted on a computer equipped with an Intel(R) Core (TM) i5-10500 1.8 GHz CPU, using the MATLAB R2022a platform.

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TABLE IV PARAMETERS FOR EACH SERVER

Parameter	Server 1	Server 2	Server 3
P_{other}	50 W	20 W	10 W
P_{idle}	40 W	10 W	5 W
a	135	55	15
b_0	675	275	85
b_1	160	150	140
c_0	30	10	3
c_1	0.2	0.08	0.05
c_2	3.3×10^{-5}	1.6×10^{-5}	1.3×10^{-5}
c_3	7.4×10^{-9}	6.4×10^{-10}	6.3×10^{-10}
k_{fan}	30	25	20
C_{cpu}	15 J/°C	10 J/°C	6 J/°C
R_{cpu}	0.04 °C/W	0.1 °C/W	0.2 °C/W
C_{hs}	355 J/°C	155 J/°C	64 J/°C
R_{hs}^0	0.26 °C/W	0.7 °C/W	1.8 °C/W
f_{fan}^{max}	1500 r/min	1250 r/min	1000 r/min
d	0.5	0.48	0.46



(a) Curves of information load arrival rate

(b) Curves of WG and PV outputs

Fig. 9. Normalized arrival rate of information load and output of WG & PV

B. *Performance of the server power consumption dynamic surrogate model*

The prerequisite for constructing the dynamic surrogate model is determining the maximum time constant τ_{max} of server power consumption. The ambient temperature is set to $T_{amb} = 25°C$. Based on the server parameters, different utilization rates U_0 are selected as stable-state points. The time step for the server is set to $\Delta\tau = 3s$. The sensitivity matrix for each server type at various utilization rates is derived using (25), as shown in Fig. 10. To determine τ_{max}, a threshold $\varsigma\%$ must be defined to specify when the influence of historical inputs becomes negligible. In this work, $\varsigma\% = 1\%$ is selected. This value follows the standard engineering convention for defining the settling time of a system [36]. According to this convention, the transient response is considered to have reached steady state once it remains within 1% of the final value. Fig. 10 shows that τ_{max} varies across server types and utilization rates. Notably, higher utilization increases fan speed, reducing the thermal resistance of the heatsink and resulting in decreased time constants. To ensure computational accuracy across utilization rates, the maximum time constants are set to 498s, 609s, and 534s for the three server types, respectively.

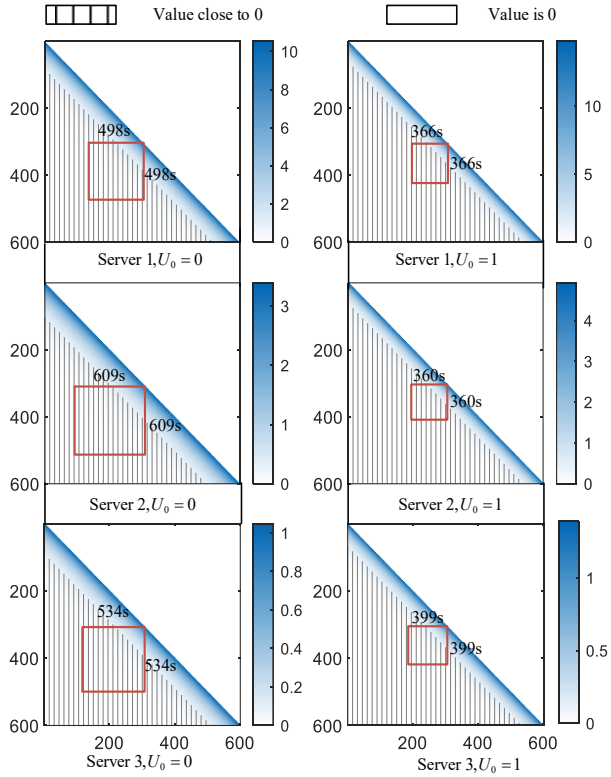


Fig. 10. Sensitivity matrix for different servers at various utilization rates

1) Performance of the ICNN-Based dynamic surrogate model

To validate the effectiveness of the proposed dynamic surrogate model, the LM, PLM, and PM models are compared. These models are considered the state-of-the-art among server power consumption models that are embeddable in optimization problems [10-12]. Based on the maximum time constant τ_{max} and the time step Δt, ICNN-based dynamic surrogate models for the three server types are developed. The results from (1)-(9) serve as the reference benchmark. The computational errors of the four models across server types are compared in Table V. As shown in Table V, the LM, PLM, and PM models exhibit larger errors due to their inability to capture the dynamic characteristics of server power consumption. In contrast, the ICNN-based model achieves a higher accuracy. These results demonstrate the effectiveness of the proposed model.

Fig. 11 further compares the computational results of the four models for Server 1. The server utilization rate exhibits a stepwise increase, with each step accompanied by fluctuations, corresponding to the two operation states shown in Fig. 2 (step changes and fluctuation changes). During step changes, the LM, PLM, and PM models fail to accurately capture the power consumption delay. During fluctuations, these models also miss the suppression process, resulting in significant variation. In contrast, the proposed dynamic surrogate model effectively captures both the delay during step changes and the suppression during fluctuations. These results demonstrate the effectiveness of the dynamic surrogate model.

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TABLE V COMPARISON OF COMPUTATIONAL ACCURACY FOR DIFFERENT MODELS

Server	Model	R^2	MAE (kW)	MAPE	RMSE (kW)
Server 1	LM	0.9812	105.4445	5.3223	126.0348
	PLM	0.9923	67.2380	3.1032	87.6288
	PM	0.9937	64.2347	2.9213	84.3246
	ICNN-based	0.9997	10.8615	0.4272	12.2996
Server 2	LM	0.9833	79.0921	6.1428	94.5769
	PLM	0.9932	44.8142	3.0928	60.5347
	PM	0.9934	43.1324	2.9832	59.4367
	ICNN-based	0.9998	9.0611	0.7207	10.4618
Server 3	LM	0.9871	37.0113	4.1195	45.7067
	PLM	0.9933	24.4785	2.5634	33.0102
	PM	0.9935	24.1237	2.4124	32.1297
	ICNN-based	0.9988	12.3422	1.0578	13.6484

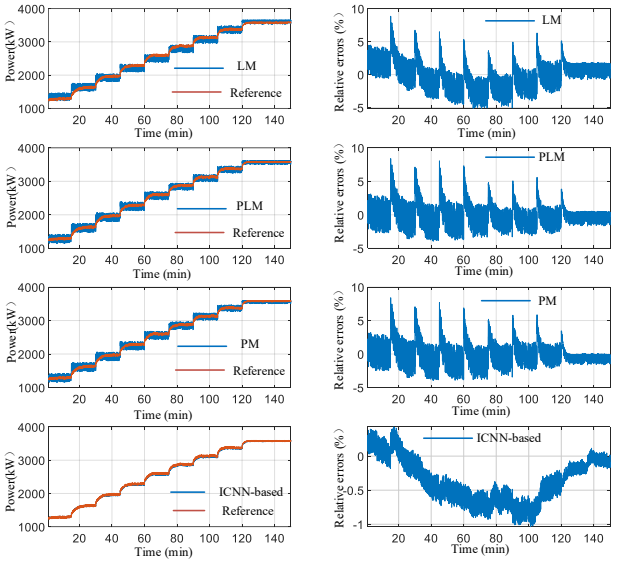


Fig. 11. Computational results of the four models for Server 1

2) Performance of the RDTP method for the dynamic surrogate model

To validate the effectiveness of the proposed RDTP method, different input temporal scopes are selected for comparison and validation. Four dynamic surrogate models for the three server types are developed, with the results from (1)-(9) serving as the reference benchmark. Table VI compares the computational accuracy across different input temporal scopes. As shown in Table VI, the models based on input time scopes Δt and $\Delta t+0.5\tau_{\max}$ exhibit significantly larger computational errors due to their incomplete consideration of historical server utilization. In contrast, the dynamic surrogate models using input time scopes $\Delta t+\tau_{\max}$ and $\Delta t+5\tau_{\max}$ achieve higher computational accuracy. These results demonstrate the ability of the proposed method to maintain model accuracy.

Fig. 12 further compares the computational results of the four dynamic surrogate models for Server 1. The model with input temporal scope Δt completely ignores the impact of historical CPU utilization, resulting in significant deviations and inaccurate power consumption calculation. The model with scope $\Delta t+0.5\tau_{\max}$ partially considers historical CPU utilization, exhibiting approximately 5% relative error. In contrast, the models with scopes $\Delta t+\tau_{\max}$ and $\Delta t+5\tau_{\max}$ fully incorporate historical inputs, limiting deviations to approximately 1%. These results validate the effectiveness of the RDTP method.

TABLE VI COMPARISON OF COMPUTATIONAL ACCURACY FOR DIFFERENT INPUT TEMPORAL SCOPES

Servers	Input temporal scope	R^2	MAE (kW)	MAPE	RMSE (kW)
Server 1	Δt	0.9527	105.7075	5.4054	155.9884
	$\Delta t+0.5\tau_{\max}$	0.9936	51.7455	2.3410	62.9652
	$\Delta t+\tau_{\max}$	0.9997	10.8615	0.4272	12.2996
	$\Delta t+5\tau_{\max}$	0.9998	9.7315	0.3965	11.8766
Server 2	Δt	0.9346	106.6674	8.4783	168.7490
	$\Delta t+0.5\tau_{\max}$	0.9868	61.5143	4.2957	75.8084
	$\Delta t+\tau_{\max}$	0.9998	9.0611	0.7207	10.4618
	$\Delta t+5\tau_{\max}$	0.9998	8.9656	0.6943	10.1069
Server 3	Δt	0.9293	51.4144	5.9520	81.9360
	$\Delta t+0.5\tau_{\max}$	0.9941	18.3693	1.9212	23.5720
	$\Delta t+\tau_{\max}$	0.9988	12.3422	1.0578	13.6484
	$\Delta t+5\tau_{\max}$	0.9991	11.6136	0.9836	12.7641

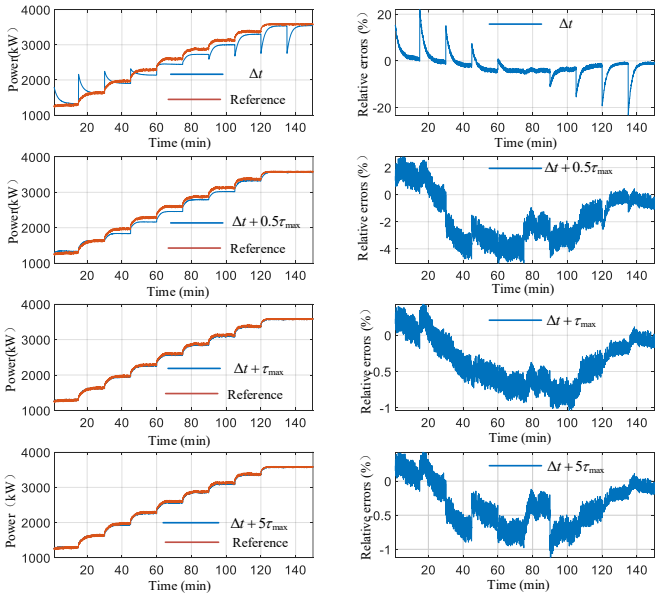


Fig. 12. Computational results of the four dynamic surrogate models for Server 1

To analyze the effectiveness of the RDTP method in reducing the dynamic surrogate model scale, Table VII compares the parameters of the four dynamic surrogate models for the three servers. The results show significant differences in the total parameter counts across different input temporal scopes. Specifically, the model with input temporal scope $\Delta t+\tau_{\max}$ has approximately 1/3 to 1/4 of the parameters compared to the model with scope $\Delta t+5\tau_{\max}$. This result demonstrates the effectiveness of the RDTP method in substantially reducing model scale.

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TABLE VII COMPARISON OF MODEL SCALE FOR DIFFERENT INPUT TEMPORAL SCOPES

Servers	Input time scope	Input	Output	Hidden layer	Parameter
Servers 1	Δt	300	300	[300 300 300]	632,400
	$\Delta t+0.5\tau_{\max}$	383	300	[383 342 300]	796,511
	$\Delta t+\tau_{\max}$	466	300	[466 383 300]	986,810
	$\Delta t+5\tau_{\max}$	1130	300	[1130 715 300]	3,506,690
Servers 2	Δt	300	300	[300 300 300]	632,400
	$\Delta t+0.5\tau_{\max}$	402	300	[402 356 300]	844,144
	$\Delta t+\tau_{\max}$	503	300	[503 402 300]	1,081,631
	$\Delta t+5\tau_{\max}$	1315	300	[1315 808 300]	4,524,511
Servers 3	Δt	300	300	[300 300 300]	632,400
	$\Delta t+0.5\tau_{\max}$	389	300	[389 345 300]	809,399
	$\Delta t+\tau_{\max}$	478	300	[478 389 300]	1,016,702
	$\Delta t+5\tau_{\max}$	1190	300	[1190 745 300]	3,821,270

3) Generalizability of the ICNN-Based dynamic surrogate model

To validate the generalizability of the proposed ICNN-based dynamic surrogate model, its computational accuracy is evaluated under different ambient temperatures, as summarized in Table VIII. The model demonstrates stable performance across the tested range from 10°C to 25°C. For all three server types, key metrics remain consistently strong with only minor variations. The results confirm that the ICNN-based dynamic surrogate model has good generalizability for ambient temperature variations.

TABLE VIII COMPARISON OF COMPUTATIONAL ACCURACY FOR DIFFERENT AMBIENT TEMPERATURES

Server	T_{amb}	R^2	MAE (kW)	MAPE	RMSE (kW)
Server 1	10°C	0.9996	11.2325	0.4419	12.7228
	15°C	0.9997	10.7526	0.4230	12.1787
	20°C	0.9998	10.4378	0.4106	11.8217
	25°C	0.9997	10.8615	0.4272	12.2996
Server 2	10°C	0.9998	9.0924	0.7232	10.4981
	15°C	0.9996	9.2673	0.7371	10.6999
	20°C	0.9997	9.1328	0.7264	10.5445
	25°C	0.9998	9.0611	0.7207	10.4618
Server 3	10°C	0.9989	11.9413	1.0234	13.2046
	15°C	0.9990	11.7523	1.0072	12.9955
	20°C	0.9987	12.6314	1.0826	13.9684
	25°C	0.9988	12.3422	1.0578	13.6484

4) Sensitivity analysis of the RDTP threshold

A sensitivity analysis is conducted to assess the influence of the RDTP threshold (ζ%) on the trade-off between model accuracy and scale, with results detailed in Table IX. The analysis reveals that a stricter threshold (e.g., 0.5%) yields a marginally higher accuracy at the cost of a significantly larger model (20-35% more parameters), while a more relaxed threshold (e.g., 2% or 5%) drastically reduces accuracy. The selected value of ζ%=1% represents the optimal compromise, maintaining modeling accuracy (e.g., MAPE below 1.06% for all servers) while effectively reduce model scale.

TABLE IX PERFORMANCE OF THE DYNAMIC SURROGATE MODELS UNDER DIFFERENT RDTP THRESHOLD VALUE

Servers	ζ%	$\tau_{\max}$ (s)	MAPE	RMSE (kW)	Parameter
Server 1	0.5%	822	0.4037	11.9832	1,276,574
	1%	498	0.4272	12.2996	986,810
	2%	423	0.7243	23.7632	927,127
	5%	324	1.5265	49.2167	850,452
Server 2	0.5%	1005	0.6937	10.0134	1,461,791
	1%	609	0.7207	10.4618	1,081,631
	2%	519	1.4325	21.8324	1,004,951
	5%	399	3.6735	60.5632	908,311
Server 3	0.5%	879	0.9876	12.9465	1,333,271
	1%	534	1.0578	13.6484	1,016,702
	2%	453	1.2138	15.7324	951,007
	5%	348	1.7134	20.3248	868,460

C. *Optimal operation results for practical system*

1) Accuracy and optimality

To validate the accuracy and optimality of the proposed method, four cases are compared:

Case 1: IDC-MG optimal operation with embedded LM.

Case 2: IDC-MG optimal operation with embedded PLM.

Case 3: IDC-MG optimal operation with embedded PM.

Case 4: IDC-MG optimal operation with embedded dynamic surrogate model (Input temporal scope $\Delta t+\tau_{\max}$).

TABLE X CASE COMPARISON (UNIT: YUAN)

Case	Value type	C_{grid}	C_{fuel}	$C_{penalty}$	C_{total}
1	Calculate value	3182	14590	672	18444
	Accurate value*	3182	14590	1662	19434
2	Calculate value	3185	14589	674	18448
	Accurate value*	3185	14589	1664	19438
3	Calculate value	3184	14588	678	18450
	Accurate value*	3184	14588	1648	19420
4	Accurate value*	3184	14590	689	18463

Table X compares the operation costs across cases. The calculated values for Cases 1-3 reflect the direct outputs of their respective models. The accurate values are derived by applying model 4 with fixed operation measures from each model. Cases 1-3 neglect the dynamic characteristics of server power consumption. While their calculated costs appear optimal, the accurate values indicate that the proposed model (Case 4) reduces costs by approximately 5.00%, 5.02%, and 4.90% in comparison to Cases 1, 2, and 3, respectively. This discrepancy occurs because ignoring server power consumption dynamics leads to significant calculation errors. These errors result in system regulation deviations, which force the tie-line transmission power to exhibit greater volatility. The increased fluctuations lead directly to higher penalty costs. These results demonstrate the effectiveness of the proposed model.

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Fig. 13 presents the power balance results of IDC-MG for Case 4. The electricity input comprises five components: the electricity transmitted by tie-line, the electricity generated by PV, the electricity generated by WT, the electricity output from MT, and the electricity discharge from EES. These power inputs collectively supply both the server load and other load within the data center.

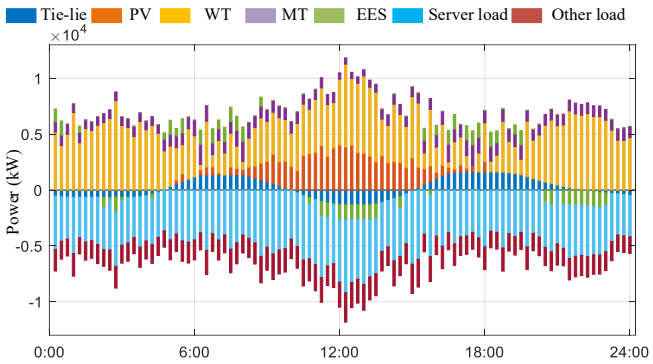


Fig. 13. Power balance results of IDC-MG for Case 4

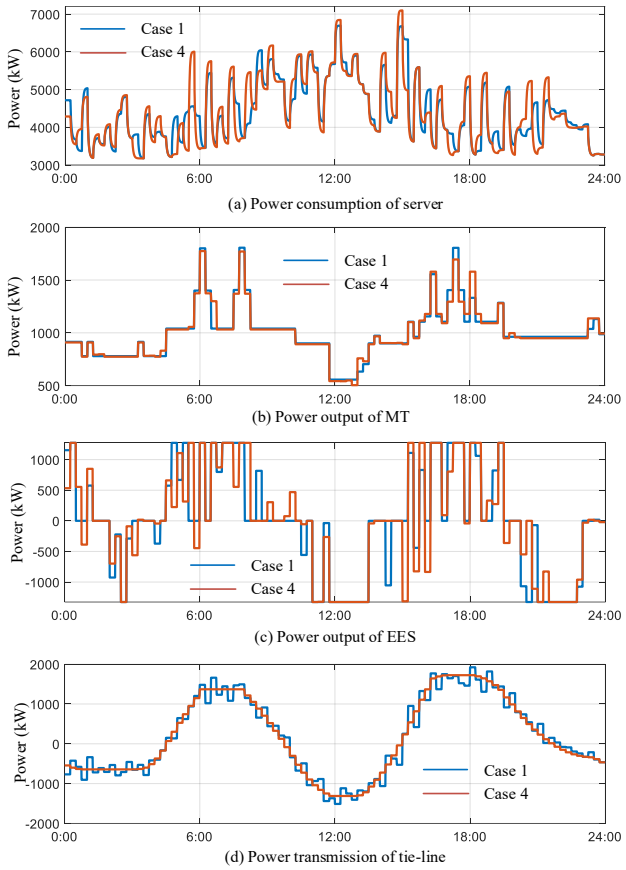


Fig. 14. Device operation results of IDC-MG for Cases 1 and 4

Fig. 14 presents the device operation results of IDC-MG for Cases 1 and 4. As shown in Fig. 14(a), the server power consumption exhibits dynamic variations, with distinct patterns influenced by the temporal shifting of the information load. In Figs. 14(b) and (c), the MT power output shows minimal differences, while the server power consumption is primarily regulated through the EES, causing significant discrepancies in the EES power output. Fig. 14(d) demonstrates differences in the tie-line transmission power, with greater fluctuations in Case 1 compared to Case 4. These results highlight the advantage of considering the dynamic characteristics of server power consumption in IDC-MG optimal operation.

2) Computational efficiency

To validate the effectiveness of the proposed model, Case 5 is designed for comparison:

Case 5: IDC-MG optimal operation with embedded dynamic surrogate model (Input temporal scope $\Delta t + 5\tau_{\max}$).

The parameters and computational times for Cases 4 and 5 are compared in Table XI. Case 4 uses 859,460 continuous variables, 338,208 integer variables, and requires 121.7 seconds of computational time. In contrast, Case 5 uses 1,491,792 continuous variables, 653,280 integer variables, and requires 404.6 seconds. The higher variable count in Case 5 increases the solution time by 282.9 seconds. The proposed method reduces problem complexity by decreasing the number of variables. This reduction improves computational efficiency by 69.9%. These results demonstrate the effectiveness of the RDTP method for IDC-MG optimal operation.

TABLE XI CASE COMPARISON

Case	Continuous variables (count)	Integer variables (count)	Solution time (s)
4	859,460	338,208	121.7
5	1,491,792	653,280	404.6

VI. EXTENDED CASE STUDY

A. *Test system description*

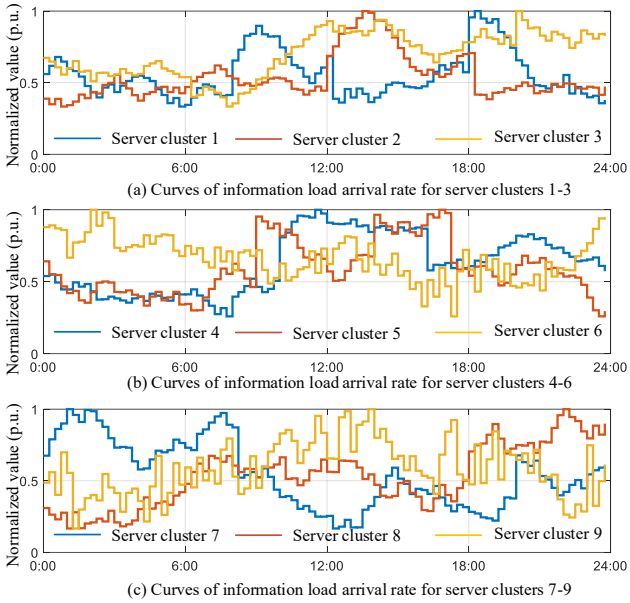
To further validate the model scalability, a test system is constructed. The system includes a WT with a 35 MW capacity, a PV with a 15 MW capacity, an MT with a 6 MW capacity and ramp rates of 1.5 MW/15 min, and an EES with a 15 MWh capacity and 4.9 MW maximum power. The data center comprises nine server clusters: Server cluster 1-3 have 5,000 high-performance servers; Server cluster 4-6 have 10,000 cloud servers; and Server cluster 7-9 have 15,000 edge servers. The server parameters are listed in Table IV, and the data center operates at a PUE of 1.3.

The upper and lower limits of the tie-line transmission power are -10 MW and 10 MW, respectively. The electricity purchase price is 0.6 yuan/kWh, and the fuel cost coefficients for the MT are $a_{\mathrm{MT}} = 0.00008$, $b_{\mathrm{MT}} = 0.4$, and $c_{\mathrm{MT}} = 0$. The power fluctuation penalty coefficients are $a_{\mathrm{penalty}} = 0.000025$ and $b_{\mathrm{penalty}} = 0.03$. The normalized arrival rate of information load is shown in Fig. 15, and output of WG & PV are shown in Fig. 9. For information loads, the real-time to delay-tolerant load ratio is 1:1, with a maximum delay for R=1 hour for delay-tolerant loads. The maximum processing rates for Servers 1, 2, and 3 are 15, 5, and 2 request/s, respectively. The peak information load arrival rates for server clusters 1-3, 4-6, and 7-9 are 70326, 31797, and 24222 request/s, respectively.

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The server utilization rates are constrained to a lower limit of 0.1 and an upper limit of 0.95.



(a) Curves of information load arrival rate for server clusters 1-3

(b) Curves of information load arrival rate for server clusters 4-6

(c) Curves of information load arrival rate for server clusters 7-9

Fig. 15. Normalized arrival rate of information load

B. *Optimal operation results for test system*

In the test system, we re-run all comparative cases defined in Section V.C. For clarity, the cases in the test system are denoted as S1 to S5, corresponding exactly to the definitions of Cases 1 to 5 in the practical system. The results, summarized in Tables XII, confirm the scalability of the proposed model. Cases S1-S3, which ignore server power dynamics, show a comparative disadvantage. The proposed model (S4) achieves cost reductions of approximately 5.45% against these cases. This consistent performance gap underscores the model robustness in preventing cost penalties across different system scales.

TABLE XII CASE COMPARISON (UNIT: YUAN)

Case	Value type	C_{grid}	C_{fuel}	$C_{penalty}$	C_{total}
S1	Calculate value	63353	15247	1349	79949
	Accurate value*	63353	15247	6016	84616
S2	Calculate value	63341	15258	1357	79956
	Accurate value*	63341	15258	6003	84602
S3	Calculate value	63348	15250	1347	79945
	Accurate value*	63348	15250	6008	84606
S4	Accurate value*	63358	15245	1395	79998

TABLE XIII CASE COMPARISON

Case	Continuous variables (count)	Integer variables (count)	Solution time (s)
S4	2,558,218	1,014,240	923.5
S5	4,455,214	1,959,456	3284.2

Furthermore, as detailed in Table XIII, the method achieves a 71.9% improvement in computational efficiency in Case S4. This gain is attained by reducing the variable set. A high-quality solution is obtained within approximately 15 minutes. These results demonstrate that the core advantages of the model are preserved in more complex systems.

VII. DISCUSSION

This section discusses the two core methodological components within the proposed IDC-MG optimal operation framework: the ICNN surrogate model and the RDTP method.

- The ICNN surrogate model exhibits the following characteristics. First, as a data-driven model, it is inherently sensitive to the training data, as its accuracy fundamentally depends on the quality and representativeness of the dataset. Second, its robustness to parameter variations is conditional, contingent on whether such variations significantly shift the data distribution. Third, it demonstrates strong generalizability across diverse hardware and environments, as validated in the case studies. Fourth, it possesses practical applicability by being trainable on noisy, real-world operational data. Fifth, its architectural convexity grants it inherent scalability. This allows convex surrogate models of numerous server clusters to be embedded concurrently as a set of compatible convex constraints within the optimization problem.
- The proposed RDTP method is defined by several key features. First, it determines the input temporal scope by theoretically deriving the dynamic response boundaries of the system. Second, it provides an objective and reproducible pruning criterion, as these boundaries are derived analytically from system parameters. Third, it is designed to maintain accuracy by fully utilizing all essential historical data within the boundaries while discarding redundant external data. Finally, it acts as a dedicated complexity reduction step, lowering the input dimension for the surrogate model and thereby facilitating the construction of compact and efficient surrogate models such as the ICNN.

In summary, the integration of the ICNN surrogate model and the RDTP method enables the proposed framework to effectively balance the requirements of dynamic modeling accuracy and computational efficiency, which is essential for realizing optimal operation of IDC-MGs.

VIII. CONCLUSION

This paper proposes a novel optimal operation method for IDC-MGs based on the dynamic surrogate model of server power consumption, enabling coordination between device-level dynamics and system-level energy management. An ICNN-based dynamic surrogate model is developed to accurately capture the nonlinear dynamics of server power consumption, avoiding the complex mathematical expressions of mechanism models. A novel RDTP method is proposed, which effectively reduces the input temporal scope of the dynamic surrogate model by identifying the dynamic response boundaries of server power consumption, reducing the model scale while maintaining computational accuracy. An optimal operation framework for IDC-MGs is developed,

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incorporating the dynamic surrogate model, which can jointly optimize energy device outputs and the temporal shifting processes of information loads, while considering the dynamic characteristics of server power consumption. The proposed method is validated using both practical and test systems. The results demonstrate operation cost reductions of approximately 5.00% and 5.45%, respectively, compared to conventional models that neglect the dynamic characteristics of server power consumption. Furthermore, the ICNN-based dynamic surrogate model achieves higher computational accuracy than LM, PLM, and PM models. Meanwhile, the RDTP method significantly reduces the dynamic surrogate model scale while maintaining computational accuracy.

Future research should tackle the uncertainty in IDC-MG source-load power through stochastic or robust optimization. The proposed modeling approach can also be applied to other equipment with complex dynamics, such as electrolyzers and fuel cells.

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